excellent additional features such as dual-side printing capabilities, direct printing from digital cameras and separate ink tanks, especially for printing photographs and the like. Thus, the entire gamut of printers can be explored, depending upon one's needs. Keeping this in mind, we tested 21 inkjet printers from leading vendors in the Indian market. We evaluated each printer for three application areas—home printing, office printing and professional printing—to help you choose the one best suited to your needs.

How we tested

Having established the areas of home, office and professional printing to evaluate a printer, we decided to further define the areas according to the printing quality offered and features available. For the home segment, the distributed weightage was 50 per cent for price, 30 per cent for performance and 20 per cent for features. A printer that gave decent printing speeds, acceptable photo printouts and affordable (in the price range of Rs 3,000 to 5,500) was considered an ideal home solution. For establishing a printer as an ideal office printer, the score was divided and price got 20 per cent, performance got 50 per cent and features, 30 per cent. In the time-saving environment of an office, speed takes paramount importance over any other factor. Features such as dual-side printing, ink status through driver, larger buffer memory, etc, were also given the due consideration. For professional printing, the overall score was split into 20 per cent for the price, 50 per cent for performance and 30 per cent for features. The only difference here was that, under performance, speed was given lower priority and the output quality more importance.

We tested all the printers on a text document, a combination document that contained both text and pictures, and a pure image document. With the benchmark score in hand, we then calculated the Suitability to Task Index (STTI) for each application area. With this STTI score in hand, we could easily define a printer's suitability to any of the three defined application areas. For example, if a printer's STTI for the home segment is 70 points, 67 for the office category, but 85 for the professional application, then it is more suited to professional applications.

All the test files, except the image file were printed for two different quality settings. The text and the combination document were printed at normal and best quality settings; the image file was printed using only the best quality settings. For printing the text and combination document, we used a 75 GSM (gram per square meter) normal paper, whereas for the image file, we used a 140 GSM photo-quality paper. These test documents incorporate every small detail that a printer needs to reproduce the smallest point text, lines and specular effect on metal.

The combination document has many sections with text of varying points; lines, circles, seven colour bars of decreasing gradient and a small picture at one corner. Each of these subsection tests the printer's ability to reproduce tiny details. The varying font-size tests the printers' ability to differentiate between adjacent fonts and reproduce the smallest possible text. Whereas the circle of lines test a printer's ability to reproduce lines without jarring the edges. Similarly, the seven colour bar tests the colour reproduction capabilities of the printer's cartridges.

The image document is made up of many elements, with varying contrast and brightness areas. This document tests the printers

1 Uniformity of the outline circle: *We checked for misalignments and missing patches*
2 Resolution test: *We examined the printer's ability to reproduce closely packed converging lines*
3 and 4 Quality of photos: *Specific details for each photo were mapped (clarity of the grille pattern, the shine on the fruit, and so on)*
5 Yellow text on black background: *This area helped us evaluate the printer's ability to print high-contrast text*
6 Number of distinct colour bars: *There are 10 bars in each colour pattern, and each bar has two shades. If a printer could print two distinct shades for each bar in every colour pattern, it was awarded a maximum score of 10*
7 Readability of fine text: *This text is in varying point sizes, ranging from 6 to 2 points. The printer was awarded one point for each readable line*

Crouching Printer, Hidden Costs

The norm, while buying peripherals, is to base purchases on their price tag. For printers, there are other significant running costs involved, primarily the cost per page.

For instance, assume that a black ink cartridge for the printer you have bought costs Rs 1,000, and can ideally print 2,000 pages. You will thus be paying 50 paisa for every page that you print. The exact cost will vary depending on the printing quality you use (Draft will be cheaper, while High-quality printouts will cost you dearly), but this is a fairly reliable method of comparing running costs across the board.

So what does this running cost mean to you? Well, if you just print a page a week, this figure means absolutely nothing. On the other hand, if your printer will be used to a great extent (several pages a day), keep the cost per page in mind. A seemingly cheap printer with a high cost-per-page figure will end up being more expensive in the long run, than a costlier printer with low running costs.